Absence of Odd-Parity Bound States in a Shallow Square Well

Source: Cambridge Quantum Mechanics, Example Sheet 1, Question 7; accompanies David Tong's notes.

Consider a square well potential with $U(x) = -U$ for $|x| < a$ and $U(x) = 0$ otherwise (U is a positive constant). Show that there are no bound states (normalisable energy eigenfunctions) which satisfy $\chi(-x) = -\chi(x)$ (i.e.\ which have odd parity) if

$$a^2 U < \frac{(\pi\hbar)^2}{8m}. \tag{1.1}$$

Solution. For the state to be bounded, we require $-U < E < 0$. Solving the TISE, we find

$$\chi(x) = \begin{cases} Ae^{\kappa x} & x \leq -a, \\ C\sin(kx) + D\cos(kx) & |x| < a, \\ Ee^{-\kappa x} & x \geq a, \end{cases} \tag{1.2}$$

with

$$\kappa = \frac{\sqrt{-2mE}}{\hbar} \qquad \text{and} \qquad k = \frac{\sqrt{2m(U+E)}}{\hbar}. \tag{1.3}$$

For the wavefunction to be odd, we require $-A = E$ and $D = 0$. Imposing the boundary conditions, we find

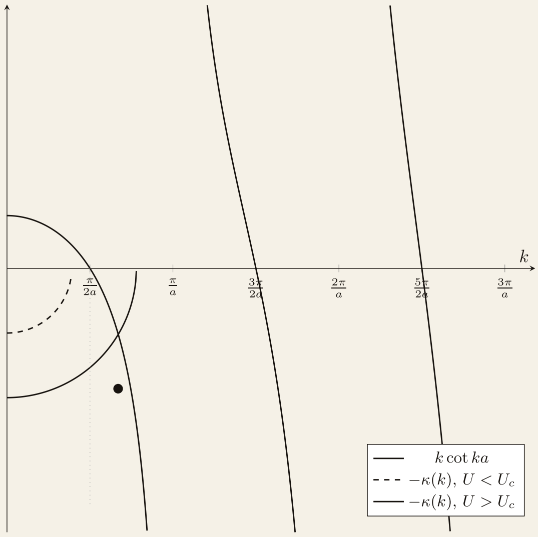
$$\frac{\tan ka}{ka} = -\frac{1}{\kappa}, \tag{1.4}$$

which is an implicit transcendental equation for E. We can rewrite the equation as

$$k \cot ka = -\kappa(k), \tag{1.5}$$

where

$$\kappa^2 + k^2 = \frac{2mU}{\hbar^2}. \tag{1.6}$$



Graphical solution of the odd-parity bound-state equation $k \cot ka = -\kappa(k)$, where $\kappa = \sqrt{2mU/\hbar^2 - k^2}$. The dashed semicircle ($U < U_c$) does not intersect $k \cot ka$ — no odd bound state exists. The solid semicircle ($U > U_c$) intersects the first branch (dot) — one odd bound state exists. The critical value is $U_c = \pi^2\hbar^2/8ma^2$, at which the circle just reaches $k = \pi/2a$ with $\kappa = 0$.

Since cot has a greater magnitude of gradient than a circle; for a solution to exist, we require the radius of the circle to be greater than $ka = \pi/2$. That is,

$$\frac{\sqrt{2mU}}{\hbar} > \frac{\pi}{2a} \implies Ua^2 > \frac{\pi^2\hbar^2}{8m}. \tag{1.7}$$